Bases and Linear Maps

Span, uniqueness, and dimension

MTH 317H · Week 8 · 75-minute review

Name: _____

How we will work

You may work independently or with others. Write your reasoning and explain why each step follows. Before calculating, name your goal and choose a tool. Afterward, check what your answer means.

1. Think, explain, revise

All vector spaces are over $\mathbb{R}$. Here $\mathcal{P}_n$ means polynomials of degree *at most* n, including the zero polynomial, and $\dim \mathcal{P}_n = n + 1$.

Task 1. Which claims survive?

Mark true or false. Work through your reasons, then revise your answer if needed. For a false statement, give a counterexample; for a true one, name the reason.

Claim	Initial answer	Revised answer
(a) Three distinct nonzero vectors in $\mathbb{R}^3$ form a basis.	F	F
(b) A linearly independent list of three vectors in $\mathcal{P}_3$ spans $\mathcal{P}_3$.	F	F
(c) If $T\colon \mathbb{R}^3 \to \mathbb{R}^3$ is linear and $\operatorname{null} T = \{0\}$, then T is surjective.	T	T

Solution. (a) False: $(1,0,0), (2,0,0), (3,0,0)$ are distinct and nonzero but dependent. (b) False: $1, x, x^2$ are independent but miss x^3; $\dim \mathcal{P}_3 = 4$. (c) True: rank–nullity gives $\dim \operatorname{range} T = 3$, so the range is all of $\mathbb{R}^3$.

Today's goals

Justify a basis; choose between definitions and dimension arguments; describe null spaces and ranges; connect injectivity to uniqueness and surjectivity to existence; use rank–nullity with its hypotheses.

2. Build a basis; compare approaches

Task 2. A list that works, and a list that almost works

Consider these lists in $\mathcal{P}_3$:

$$\mathcal{B} = (1+x,\ 1-x,\ x^2,\ x^3), \qquad \mathcal{C} = (1+x,\ 1-x,\ 2,\ x^2).$$

Plan first: what does it take to prove that a list is a basis?

(a) Prove that $\mathcal{B}$ is linearly independent. Start with a linear combination equal to 0.

> **Solution.** If $a(1+x)+b(1-x)+cx^2+dx^3 = 0$, coefficient comparison gives $a+b=0$, $a-b=0$, $c=d=0$. Thus $a=b=c=d=0$, so $\mathcal{B}$ is independent.

(b) Prove that $\mathcal{B}$ spans by expressing an arbitrary $q = q_0+q_1x+q_2x^2+q_3x^3$ using $\mathcal{B}$. Then use a dimension theorem to finish the basis proof from (a). Compare the approaches: what work does the theorem save, and what information does the direct method give?

> **Solution.** Directly,
>
> $$q = \frac{q_0+q_1}{2}(1+x) + \frac{q_0-q_1}{2}(1-x) + q_2x^2 + q_3x^3.$$
>
> This supplies coordinates and proves spanning. Alternatively, $\mathcal{B}$ is an independent list of length $4 = \dim \mathcal{P}_3$, so it is a basis. The theorem saves a separate spanning computation; it does not supply the coordinates.

(c) Diagnose $\mathcal{C}$: exhibit a nontrivial dependence relation, identify its span and dimension, and replace exactly one entry to obtain a basis of $\mathcal{P}_3$. Explain why your repair works.

> **Solution.** $(1+x)+(1-x)-2=0$ is nontrivial. The first two entries generate 1 and x, so $\operatorname{span}\mathcal{C} = \mathcal{P}_2$ and its dimension is 3. Replacing 2 with x^3 gives a reordering of $\mathcal{B}$, hence a basis. Four distinct nonzero entries were not enough.

Checkpoint. Say which *space* each dimension refers to. The length of a list and the dimension of its span need not agree.

3. Two measurements of a polynomial

Task 3. What can two values tell us?

Let $E\colon \mathcal{P}_2 \to \mathbb{R}^2$ be defined by $E(p) = (p(1), p(0))$. Predict whether two values determine a quadratic uniquely. Then investigate.

(a) Prove linearity without expanding a general quadratic: compare $E(\alpha p+\beta q)$ with $\alpha E(p)+\beta E(q)$ for arbitrary p, q, α, β.

> **Solution.** Evaluation distributes over linear combinations:
>
> $$\begin{aligned} E(\alpha p + \beta q) &= (\alpha p(1) + \beta q(1), \alpha p(0) + \beta q(0)) \\ &= \alpha E(p) + \beta E(q). \end{aligned}$$
>
> Thus E is linear.

(b) Find $\operatorname{null} E$, a basis for it, and its dimension. One approach uses coefficients; another uses the roots 0 and 1. Compare them. Is E injective?

> **Solution.** For $p = a + bx + cx^2$, $E(p) = 0$ means $a = 0$ and $b + c = 0$. Thus $p = c(x^2 - x)$. Equivalently, a polynomial vanishing at $0, 1$ is divisible by $x(x-1)$; the degree bound makes the quotient constant. Hence $\operatorname{null} E = \operatorname{span}(x^2 - x)$, with basis $(x^2 - x)$ and dimension 1. It is not injective: $E(x^2 - x) = E(0)$.

(c) Start with an arbitrary target $(r, s) \in \mathbb{R}^2$. Construct p with $E(p) = (r, s)$. State the range, a basis for the range, and whether E is surjective.

> **Solution.** Choose $p_0 = s+(r-s)x$. Then $p_0(1) = r$ and $p_0(0) = s$. Thus $\operatorname{range} E = \mathbb{R}^2$, with basis $((1, 0), (0, 1))$, dimension 2, and E is surjective. Notice the order of the coordinates: the constant term is s, not r.

(d) Find *all* polynomials sent to (r, s). Explain why your answer contains every solution, and use it to describe both existence and uniqueness.

> **Solution.** All solutions are $s + (r - s)x + t(x^2 - x)$, $t \in \mathbb{R}$. Every such polynomial has the required values. Conversely, if $E(p) = (r, s) = E(p_0)$, linearity gives $p - p_0 \in \operatorname{null} E$. Every target is attained, but no target has a unique preimage.

Check: $\dim \mathcal{P}_2 = \dim \operatorname{null} E + \dim \operatorname{range} E$ becomes 3=1+2.

4. A list viewed as a linear map

Task 4. Where do the coefficients go?

Let $v_1 = (1, 0, 1)$, $v_2 = (0, 1, 1)$, $v_3 = (1, 1, 2)$ and define

$$S\colon \mathbb{R}^3 \longrightarrow \mathbb{R}^3, \qquad S(a, b, c) = av_1 + bv_2 + cv_3 = (a + c, b + c, a + b + 2c).$$

(a) Predict a dependence relation among the v_i. Explain briefly why S is linear, then find $\operatorname{null} S$, a basis for it, and its dimension. Translate a nonzero null vector back into a dependence relation.

Solution. $v_3 = v_1 + v_2$. Distributing coefficients in a linear combination shows that S is linear. $S(a, b, c) = 0$ forces $a = b = -c$, hence $\operatorname{null} S = \operatorname{span}((-1, -1, 1))$, with dimension 1. The generator gives $-v_1 - v_2 + v_3 = 0$.

(b) Prove, in both directions, that $\operatorname{range} S = U := \{(u, v, w) \in \mathbb{R}^3 : w = u + v\}$. Find a basis and dimension of U. Is S injective? Surjective?

Solution. Every output has third coordinate equal to the sum of the first two, so $\operatorname{range} S \subseteq U$. Conversely, $(u, v, u + v) = S(u, v, 0)$, so $U \subseteq \operatorname{range} S$. The list (v_1, v_2) spans U and is independent by its first two coordinates; thus $\dim U = 2$. S is not injective, since its null space is nontrivial, and not surjective onto $\mathbb{R}^3$, since $(0, 0, 1) \notin U$.

(c) Keep the formula but change the codomain to U: $\widetilde{S}\colon \mathbb{R}^3 \to U$. Which of the following change: null space, range, injectivity, surjectivity? Explain.

Solution. The null space, range, and non-injectivity are unchanged. But $\widetilde{S}$ is surjective because its codomain equals its range U.

Complete the dictionary, then justify one row

For a list $w_1, \ldots, w_k$ in V, let $F\colon \mathbb{R}^k \to V$ send coefficients to $\sum a_i w_i$.

The list is independent exactly when F is injective.

The list spans V exactly when F is surjective.

The list is a basis of V exactly when F is bijective.

Independence means the only coefficients giving 0 are all zero. Spanning means every target occurs. Both together mean each target has unique coefficients.

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5. What dimensions force

Rank–nullity: the Fundamental Theorem of Linear Maps
If $T\colon V \to W$ is linear and V is finite-dimensional, then

$$\dim V = \underbrace{\dim \operatorname{null} T}_{\text{nullity } T} + \underbrace{\dim \operatorname{range} T}_{\text{rank } T}.$$

Also T is injective iff $\operatorname{null} T = \{0\}$; it is surjective iff $\operatorname{range} T = W$. If W is finite-dimensional, surjectivity is equivalent to $\operatorname{rank} T = \dim W$.

Task 5. Possible, impossible, or not determined?

In each row T is linear. Fill every cell. Write "not determined" when appropriate and justify it with two examples. For an impossible row, identify the contradiction.

	Given	Nullity	Rank	Injective?	Onto?
A	$\mathbb{R}^4 \to \mathbb{R}^3$, rank 3	1	3	No	Yes
B	$\mathbb{R}^3 \to \mathbb{R}^4$, nullity 0	0	3	Yes	No
C	$\mathbb{R}^3 \to \mathbb{R}^3$, rank 2	1	2	No	No
D	$\mathbb{R}^4 \to \mathbb{R}^2$, nullity 1	–	–	–	–
E	$\mathbb{R}^4 \to \mathbb{R}^3$, no further data	1–4	0–3	No	Not determined

Solution. A–C follow from rank–nullity and comparison with the codomain dimension. D is impossible: rank would be $4 - 1 = 3 > 2$. In E, rank is an integer $0 \le r \le 3$ and nullity is $4 - r$. The zero map has rank 0, nullity 4, and is not onto; $(a, b, c, d) \mapsto (a, b, c)$ has rank 3, nullity 1, and is onto. No such map is injective.

Task 6. Turn the pattern into a proof

Suppose $\dim V = \dim W = n < \infty$ and $T\colon V \to W$ is linear. Prove that T is injective if and only if it is surjective. Write the first and last lines before filling the middle; mark where each hypothesis is used.

Solution. By linearity, injectivity is equivalent to $\operatorname{null} T = \{0\}$, hence to nullity 0. Rank–nullity (finite-dimensional domain) makes this equivalent to rank n. Since $\operatorname{range} T$ is a subspace of the n-dimensional codomain, rank n is equivalent to $\operatorname{range} T = W$, hence to surjectivity. Each equivalence works in both directions.

Check your theorem use. Rank counts independent output directions, not the number of coordinates written in the output. The sum in rank–nullity is the dimension of the *domain*.

6. Audit an argument

Task 7. Proof clinic

For each argument: give a verdict, identify the first unjustified step, and repair the reasoning. State whether the original claim is true or false.

A. "If (b_1, b_2, b_3) is a basis of V and $T\colon V \to W$ is linear, then $(T(b_1), T(b_2), T(b_3))$ is a basis of range T. Indeed, every output is a combination of these three vectors, so they form a basis and the rank is 3."

> **Solution.** The first error is "so they form a basis": spanning does not imply independence. The claim is false; the zero map sends all three basis vectors to 0 and has rank 0. Repair: the images span the range; reduce this list to a basis. The original image list is a basis of the range if T is injective.

B. "For E in Task 3, the null space is $\operatorname{span}(x^2 - x)$, which has dimension 1. The codomain has dimension 2, so rank–nullity gives rank $= 2 - 1 = 1$. Therefore E is not surjective."

> **Solution.** The first error uses the codomain dimension in rank–nullity. The domain $\mathcal{P}_2$ has dimension 3, so rank $= 3 - 1 = 2$. Thus E is surjective, and the conclusion is false.

Explain: what extra evidence turns "these vectors span" into "these vectors form a basis"? Independence, or a spanning list of length equal to the known dimension.

Extension A. Why rank–nullity works

Optional

Task 9. Build a proof from a basis

Let $T\colon V \to W$ be linear, with $\dim V = n < \infty$. Start with a basis $(u_1, \dots, u_k)$ of $\operatorname{null} T$ and extend it to a basis

$$(u_1, \dots, u_k, v_1, \dots, v_{n-k}) \quad \text{of } V.$$

Prove that $(T(v_1), \dots, T(v_{n-k}))$ is a basis of $\operatorname{range} T$. Use the basis-extension theorem, but do not use rank–nullity to prove itself.

(a) Prove that these images span the range. Begin with an arbitrary $w \in \operatorname{range} T$; explain why there is an input to expand in the displayed basis.

> **Solution.** Let $w \in \operatorname{range} T$. Choose $z \in V$ with $T(z) = w$ and write $z = \sum_{i=1}^{k} a_i u_i + \sum_{j=1}^{n-k} b_j v_j$. Applying T kills the u_i terms, giving $w = \sum_j b_j T(v_j)$. Conversely, any combination of these images equals $T(\sum_j b_j v_j)$ and is in the range. Thus they span $\operatorname{range} T$.

(b) Prove independence. If $\sum_j c_j T(v_j) = 0$, where does $\sum_j c_j v_j$ lie? Why must all c_j be zero?

> **Solution.** Linearity gives $T(\sum_j c_j v_j) = 0$, so $\sum_j c_j v_j \in \operatorname{null} T$. Write $\sum_j c_j v_j = \sum_i d_i u_i$. Hence $\sum_j c_j v_j - \sum_i d_i u_i = 0$. Independence of the extended basis forces every c_j and d_i to be zero. Therefore the image list is independent.

(c) Count the two basis lengths to obtain rank–nullity. Explain what happens when $T = 0$.

> **Solution.** Nullity is k and rank is $n - k$, so their sum is $n = \dim V$. If $T = 0$, then $k = n$ and the image list is empty: the empty list is a basis of the zero range, which has dimension 0. There is no basis vector "0" to count.

Check your reasoning. In which step would the argument fail if we applied T to an arbitrary basis of V and simply claimed that all its images were independent? Kernel directions map to 0. Independence above relies on separating a kernel basis from the remaining vectors before taking images.

Extension B. A null space of functions

Optional

Task 10. One more constraint

Let $\mathcal{C}^2(\mathbb{R})$ denote twice continuously differentiable real-valued functions, and set

$$D = \{y \in \mathcal{C}^2(\mathbb{R}) : y'' - 7y' + 10y = 0\}, \qquad H = \{y \in D : y'(0) = 0\}.$$

Given fact from differential equations: every $y \in D$ has the form $y = c_1 e^{2x} + c_2 e^{5x}$. You do not need to prove this fact.

(a) Define a linear map $A\colon \mathcal{C}^2(\mathbb{R}) \to \mathcal{C}^0(\mathbb{R})$ whose null space is D. Explain why D is a subspace. Verify that e^{2x} and e^{5x} lie in it.

> **Solution.** Take $A(y) = y'' - 7y' + 10y$. Differentiation and linear combinations are linear, so A is linear and $D = \operatorname{null} A$ is a subspace. For e^{rx}, $A(e^{rx}) = (r^2 - 7r + 10)e^{rx} = (r-2)(r-5)e^{rx}$, which is zero for $r = 2, 5$.

(b) Prove that (e^{2x}, e^{5x}) is independent and deduce $\dim D$. Explain why the verification in (a), by itself, would not determine $\dim D$.

> **Solution.** If $ae^{2x} + be^{5x} = 0$ as functions, evaluation at 0 gives $a + b = 0$; differentiation and evaluation at 0 gives $2a + 5b = 0$. Thus $a = b = 0$. The given fact supplies spanning, so the list is a basis and $\dim D = 2$. Part (a) only proves the span is contained in D; it does not rule out other solutions.

(c) Let $\varphi\colon D \to \mathbb{R}$ be $\varphi(y) = y'(0)$. Show it is linear and surjective. Use rank–nullity to find $\dim H$, then find and verify a basis for H.

> **Solution.** Differentiation and evaluation are linear. For any $t \in \mathbb{R}$, $y = (t/2)e^{2x} \in D$ has $\varphi(y) = t$, so φ is surjective and has rank 1. Since $H = \operatorname{null} \varphi$, $\dim H = 2 - 1 = 1$. The condition is $2c_1 + 5c_2 = 0$, hence $H = \operatorname{span}(5e^{2x} - 2e^{5x})$. This generator is nonzero (its value at 0 is 3) and its derivative at 0 is $10 - 10 = 0$, so it is a basis of H.

Scope check. Apply the finite-dimensional rank–nullity theorem to $\varphi\colon D \to \mathbb{R}$; the domain of A is infinite-dimensional.

Exit ticket

Name: _____

Complete this page independently. Tear it off and hand it in at the end of class.

Task 8. Transfer the ideas

Let $L\colon \mathcal{P}_3 \to \mathbb{R}^2$ be linear with $\operatorname{null} L = \operatorname{span}(1, x)$.

(a) Find its nullity and rank. Is L injective? Surjective? Give reasons.

> **Solution.** $1, x$ are independent, so nullity $= 2$ and rank $= 4 - 2 = 2$. Not injective (nonzero kernel); surjective (range dimension equals codomain dimension).

(b) Must $L(1), L(x), L(x^2), L(x^3)$ be independent? Explain.

> **Solution.** No: $L(1) = L(x) = 0$, so the list contains zero vectors and is dependent.

Reflection

One strategy I can now explain is

One question I still have is
